

```

1 function runBatchInspection(ds)
2 % Batch image classifier
3
4 % retrieve test set size
5 numFiles = numel(ds.Files);
6
7 % 1. Initialize arrays to store true and predicted labels
8 trueLabels = categorical(NaN(1, numFiles));
9 predictedLabels = categorical(NaN(1, numFiles));
10
11 % 2. Batch Processing Loop
12 for i = 1:numFiles
13
14     % Load or generate test data for the current batch
15     img = readimage(ds, i);
16
17     % Extract ground truth and pass the data to the tubeClassifierNet model
18     actual = ds.Labels(i);
19     imgData = singleImageInspectionFunction(img, true);
20
21     % Append results to your master lists
22     trueLabels(i) = actual;
23     predictedLabels(i) = imgData.AIClass;
24
25 end
26
27 % 3. Generate Confusion Chart
28 figure;
29 cm = confusionchart(trueLabels, predictedLabels);
30 cm.Title = 'Batch Test Confusion Matrix';
31 % cm.RowDisplayLabels = {'Color Defect', 'Length Defect', 'Good', 'Metal Defect'};
32 % cm.ColumnDisplayLabels = {'Color D.', 'Length D.', 'Good', 'Metal D.'};
33 labels = string(unique(trueLabels));
34 cm.RowDisplayLabels = labels;
35 cm.ColumnDisplayLabels = labels;
36 cm.ColumnTickLabelRotation = 270;
37 cm.RowSummary = 'row-normalized'; % Shows class-specific true positive rates
38
39 % true = pass, false = fail
40 truePF = (trueLabels == 'good');
41 predictedPF = (predictedLabels == 'good');
42
43 figure;
44 cmPF = confusionchart(truePF, predictedPF);
45 cmPF.Title = 'Batch Test Confusion Matrix PF';
46 cmPF.RowDisplayLabels = {'Fail', 'Pass'};
47 cmPF.ColumnDisplayLabels = {'Fail', 'Pass'};
48
49 % Yield Rates (Actual + Predicted)
50 trueYield = (sum(truePF) / numFiles) * 100;
51 predictedYield = (sum(predictedPF) / numFiles) * 100;
52 out = sprintf("Actual Yield: %.1f%%\nPredicted Yield %.1f%%\n", ...
53     trueYield, predictedYield);
54 disp(out);
55
56
57 % Defect Rates
58 trueDR = 100 - trueYield;
59 escapedDefects = (predictedPF == true & truePF == false);
60 escapedDR = sum(escapedDefects); % Percentage of defects missed by the detection system
61 out = sprintf("Defect Rate: %.1f%%\nEscaped Defect Rate %.1f%%\n", ...

```

```
62         trueDR, escapedDR);  
63 disp(out);  
64 end
```